Computability theoretic forcing*

How to prove theorems about Turing reducibility the hard way.

Dino Rossegger

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1 Reducibility, hierarchies, and complete sets

The concept of relative computability goes back to Turing's 1939 paper and is usually modeled using oracle Turing machines. Informally a function $f : \omega \to \omega$ is **computable relative to** $A \subseteq \omega$ if there is an algorithm for computing $f(n)$ where the algorithm has access to elementary information about A, i.e., it may ask questions of the form "is $k \in A$"? during execution (and will also receive answers to these questions).

More formally, an oracle Turing machine Γ is a Turing machine with an additional read-only "oracle tape". The machine has an independent oracle reading head which it may move along the oracle tape and read the bit under it.For an infinite binary string $x \in 2^\omega$, we write Γ^x for the oracle Turing machine with x written on its oracle tape.

A function $f : \omega \to \omega$ is computable relative to $A \subseteq \omega$ if there is (a program for) an oracle Turing machine Γ such that for every n, $f(n) = \Gamma^A(n)$ (strictly speaking we put the characteristic function χ_A of A on the oracle tape and not A. To ease notation we identify A with its characteristic function when appropriate). A set $A \subseteq \omega$ is computable relative to $B \subseteq \omega$ if its characteristic function χ_A is computable relative to B.

Any computation $\Gamma^A(n)$ must terminate after a finite amount of steps and thus can only read a finite amount of the oracle tape. So there must be some $k \in \omega$ so that if $B \upharpoonright k = A \upharpoonright k$, then $\Gamma^B(n) = \Gamma^A(n)$. The least such k is called the **use** $u(\Gamma, A, n)$ of $\Gamma^A(n)$. Motivated by this we also allow finite strings $\sigma \in 2^{<\omega}$ on the oracle tape. We use the convention that if $\Gamma^\sigma(n)$ tries to move past σ on the oracle tape then $\Gamma^\sigma(n) \uparrow$ (i.e., is undefined) and that if $\Gamma^\sigma(n) \downarrow$, then it does so within $|\sigma|$ steps, i.e. $\Gamma^\sigma(n) \downarrow$ iff $\Gamma^\sigma_{|\sigma|}(n) \downarrow$.

The fact alluded to above that if $\Gamma^\sigma(n) \downarrow$ and $A \succ \sigma$, then $\Gamma^A(n) = \Gamma^\sigma(n)$ is called the **use principle**. The use principle is an important property of oracle computations. We will now gather some more facts which follow from easy adaptations of the proofs of similar facts for Turing machines.

Proposition 1.1.

1. *There is a* ***universal oracle Turing machine****, i.e., a Turing machine* Φ *such that for each Turing machine* Γ*, there is an* $e \in \omega$ *such that for all* n *and* $X \subseteq \omega$ $\Phi^X(e, n) = \Gamma^X(n)$*. We will write* Φ_e *for* $\Phi^-(e, -)$ *and refer to* $(\Phi_e)_{e\in\omega}$ *as* ***standard enumeration*** *of Turing machines.*

2. *(*The s-m-n theorem*) There is an injective computable function* $s : \omega^2 \to \omega$ *so that for all* x, y, z *and all* $X \subseteq \omega$ $\Phi^X_{s(x,y)}(z) = \Phi^X_x(y, z)$*.*

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3. *(*The padding lemma*) There is an injective computable function $p : \omega^2 \to \omega$ so that for every n and every i $\Phi_n^X = \Phi_{p(n,i)}^X$.*

4. *(*The recursion theorem*) Let $f : \omega \to \omega$ be a computable function. Then there is some e such that for all X, $\Phi_e^X = \Phi_{f(e)}^X$.*

1.1 Turing reducibility and the Turing degrees

Definition 1.1. Let $(\Phi_e)_{e\in\omega}$ be an enumeration of the oracle Turing machines. We say that A is computable relative to B, or, A is **Turing reducible to** B, $A \leq_T B$, if there is an e such that $\Phi_e^B = A$. A and B are **Turing equivalent**, $A \equiv_T B$, if $A \leq_T B$ and $B \leq_T A$.

The **Turing degree** of A is

$$deg_T(A) = \mathbf{a} = \{B : B \equiv_T A\}.$$

The set of Turing degrees is denoted by $\mathcal{D}_T$. It is partially ordered by the relation

$$\mathbf{a} \leq \mathbf{b} \iff \exists A \in \mathbf{a} \exists B \in \mathbf{b}\ A \leq_T B$$

Clearly there is a least Turing degree consisting of the computable sets, the degree of $\emptyset$, denoted by $\mathbf{0}$.

There are many other reducibilities between sets studied in computability theory such as arithmetic, hyperarithmetic, enumeration, (weak) truth table reducibility, etc. Two very important ones that will be useful to us are the following.

Definition 1.2. Let $A, B \subseteq \omega$, then

1. A is **many-one reducible to** B, $A \leq_m B$, if there is a computable function $f : \omega \to \omega$ so that $n \in A$ iff $f(n) \in B$. The induced many-one equivalence is denoted by $\equiv_m$.

2. A is **one-one reducible to** B, $A \leq_1 B$, if there is a computable injective function $f : \omega \to \omega$ so that $n \in A$ iff $f(n) \in B$. The induced one-one equivalence is denoted by $\equiv_1$.

It is not hard to see that $\leq_1$ and $\leq_m$ are finer than $\leq_T$, i.e. $A \leq_m B$ implies $A \leq_T B$. One-one equivalence is often denoted as the *recursive isomorphism* relation as Myhill's isomorphism theorem shows that $A \equiv_1 B$ if and only if there is a computable bijection $f : \omega \to \omega$ such that $n \in A$ iff $f(n) \in B$.

The **computable join** of $A, B \subseteq \omega$ is

$$A \oplus B = \{2n : n \in A\} \cup \{2n + 1 : n \in B\}.$$

Proposition 1.2. *The Turing degrees $(\mathcal{D}_T, \leq)$ form an upper semilattice with*

$$\mathbf{a} \vee \mathbf{b} = deg_T(A \oplus B).$$

Proof. Fix $A, B \subseteq \omega$ of degree $\mathbf{a}$, respectively $\mathbf{b}$. The functions $n \mapsto 2n$ and $n \mapsto 2n + 1$ are computable and thus $A, B \leq_m A \oplus B$, so $deg_T(A \oplus B) \geq \mathbf{a}, \mathbf{b}$. Suppose $\mathbf{c} \geq \mathbf{a}, \mathbf{b}$. Then for any $C \in \mathbf{c}$, $C \geq_T A, B$ and thus also $C \geq_T A \oplus B$. To see this suppose that $\Phi_{e_0}^C = A$ and $\Phi_{e_1}^C = B$ and consider the function $f : 2^\omega \times \omega \to 2$

$$f(X, n) = \begin{cases} \Phi_{e_0}^X(k/2) & if\, n = 2k \\ \Phi_{e_1}^X(k/2) & if\, n = 2k + 1 \end{cases}$$

This function is clearly computable with an index depending only on e_1, e_2 and thus there is an oracle Turing machine Φ such that $\Phi^X = f(X, -)$. To be more precise, consider the program running program e_0 on even inputs and program e_1 on odd inputs and appeal to the Church-Turing thesis. □

1.1.1 The Turing jump

The most important function between sets in computability theory is the **Turing jump**

$$\mathsf{J}: \begin{array}{l} 2^\omega \to 2^\omega \\ A \mapsto A' = \{n : \Phi_n^A(n)\downarrow\} \end{array}.$$

Theorem 1.3 (Properness of the Turing jump)**.** *For all $X \subseteq \omega$, $X <_T X'$.*

Proof. We first show that $X' \geq_T X$. For each n, construct a program $\Phi^X_{f(n)}$ which halts on any input m if and only if $n \in X$. Then $X \leq_1 X'$ via the computable function f, as $n \in X$ if and only if $\Phi^X_{f(n)}(f(n))\downarrow$ if and only if $f(n) \in X'$.

That $X \not\geq_T X'$ is the straight-forward relativization of the fact that the halting problem is not computable. We give it here to illustrate how many facts about computable functions relativize.

Suppose towards a contradiction that $X' \leq_T X$, then the partial function

$$f(n) = \begin{cases} 1 & n \notin X' \\ \uparrow & n \in X' \end{cases}$$

is partial computable from X, i.e., there is an index e such that $\Phi_e^X = f$. But then

$$\Phi_e^X(e)\downarrow \iff e \notin X' \iff \Phi_e^X(e)\uparrow,$$

a clear contradiction. □

Next we see that the Turing jump is degree invariant in a strong sense and thus can be viewed as a function on the Turing degrees.

Theorem 1.4. *For all $A, B \subseteq \omega$, $A \leq_T B \iff A' \leq_1 B'$.*

Proof. Suppose that $A \leq_T B$, via Φ_e and for each n construct a program $\Phi^B_{f(n)}$ such that for any m

$$\Phi^B_{f(n)}(m)\downarrow \iff \Phi_n^{\Phi_e^B}(n)\downarrow.$$

Then $n \in A'$ iff $\Phi^B_{f(n)}(f(n))\downarrow$ iff $f(n) \in B'$, showing that $A' \leq_1 B'$.

On the other hand assume that $A' \leq_1 B'$ via f. Note that relativizing the completeness of $\emptyset'$, if $C \leq_1 B'$, then C is c.e. in B. Hence, A is c.e. in B. To finish we need to show that $\omega \setminus A$ is c.e. in B. To see this consider g such that for all m $\Phi^A_{g(n)}(m)\downarrow$ if and only if $n \notin A$. Then $n \notin A$ if and only if $g(n) \in A'$ if and only if $f(g(n)) \in B'$. As both g and f are computable, $\omega \setminus A$ is c.e. in B. □

1.2 The arithmetical hierarchy

Definition 1.3 (The relativized arithmetical hierarchy)**.** Let $X \subseteq \omega$ and $A \subseteq \omega$. Then for all $n \in \omega$

1. $\Delta^0_0(X) = \Sigma^0_0(X) = \Pi^0_0(X) = \{A \subseteq \omega : A \leq_T X\}$

2. A is $\Sigma^0_n(X)$ if there is an X-computable predicate R such that

$$x \in A \iff (\exists \vec{u}_1)(\forall \vec{u}_2)(\exists \vec{u}_3) \cdots (Q_n \vec{u}_n)\, R(x, \vec{u}_1, \ldots, \vec{u}_n),$$

where $Q_n = \exists$ if n is odd and $Q_n = \forall$ otherwise.

3. A is $\Pi^0_n(X)$ if there is an X-computable predicate R such that

$$x \in A \iff (\forall \vec{u}_1)(\exists \vec{u}_2)(\forall \vec{u}_3) \cdots (Q_n \vec{u}_n)\, R(x, \vec{u}_1, \ldots, \vec{u}_n),$$

where $Q_n = \exists$ if n is even and $Q_n = \forall$ otherwise.

4. A is $\Delta^0_n(X)$ if and only if $A \in \Sigma^0_n(X) \cap \Pi^0_n(X)$

We write Γ^0_n instead of $\Gamma^0_n(\emptyset)$ for $\Gamma \in \{\Sigma, \Pi, \Delta\}$.

The term arithmetical hierarchy comes from the fact that the Σ^0_n sets coincide with the sets definable in $(\mathcal{N}, 0, 1, +, \cdot)$ via $\exists_n$-formulas.

Proposition 1.5. *A set $A \subseteq \omega$ is Π^0_n if and only if $\omega \setminus A$ is Σ^0_n.*

We will now connect the arithmetic hierarchy with Turing reducibility and the Turing jump. Towards this note that we can iterate the Turing jump. Given $X \subseteq \omega$, let $X^{(0)} = X$ and for $n \in \omega$ let the **$(n+1)$-th Turing jump** of X be $X^{(n+1)} = (X^{(n)})'$.

Lemma 1.6. *For every $A, X \subseteq \omega$, $A \in \Sigma^0_1(X)$ iff A is c.e. in X if and only if $A \leq_m X'$.*

Proof. Proofs before, mutatis mutandis. □

Theorem 1.7 (Post's theorem)**.** *For every $A, X \subseteq \omega$ and $n > 0$, $A \in \Sigma^0_n(X)$ iff A is c.e. in $X^{(n-1)}$ iff $A \leq_m X^{(n)}$.*

Proof. The proof is by induction with the base case covered by Lemma 1.6. That A is c.e. in $X^{(n-1)}$ iff $A \leq_m X^{(n)}$ follows by replacing X in Lemma 1.6 with $X^{(n-1)}$.

Suppose by induction that the theorem holds for n, and suppose that A is $\Sigma^0_{n+1}(X)$ defined by the computable relation R. Then the set $P = \{(n, \vec{u}_1) : \forall \vec{u}_2 \ldots Q_n \vec{u}_n R(-)\}$ is $\Pi^0_n(X)$ and note that $A \in \Sigma^0_1(P)$. As $P \in \Pi^0_n(X)$, $\omega \setminus P$ is $\Sigma^0_n(X)$ and by the induction hypothesis, $\omega \setminus P \leq_m X^{(n)}$. In particular, $P \leq_T X^{(n)}$, and as A is c.e. in P, A is also c.e. in $X^{(n)}$.

On the other hand, suppose that A is c.e. in $X^{(n)}$. Then by the base case A is $\Sigma^0_1(X^{(n)})$, i.e. $x \in A \iff \exists u R(x, u)$ for R $X^{(n)}$-computable. Clearly $X^{(n)} \leq_m X^{(n)}$ and so by hypothesis $X^{(n)}$ is $\Sigma^0_n(X)$. Let Φ_e be such that $\Phi_e^X = R$ and let σ_k be the kth finite binary string in the lexicographic ordering. We claim that

$$x \in A \iff \exists u \exists k\ \Phi_e^{\sigma_k}(x, u) \downarrow = 1 \wedge (\forall m < |\sigma_k|)(\sigma_k(m) = 1 \leftrightarrow m \in X^{(n)}).$$

To see this recall the use principle. $\Phi_e^{X^{(n)}}(x, u) \downarrow = 1$ iff there is a $\sigma_k \prec X^{(n)}$ such that $\Phi_e^{\sigma_k}(x, u) \downarrow = 1$. $(\forall m < |\sigma_k|)\sigma_n(m) = 1 \leftrightarrow m \in X^{(n)}$ is a $\Sigma^0_{n+1}(X)$-definition, as $\Sigma^0_{n+1}(X) \supset \Sigma^0_n(X) \cup \Pi^0_n(X)$. Thus, the whole definition is $\Sigma^0_{n+1}(X)$. □

Corollary 1.8. *For $X, A \subseteq \omega$, A is $\Delta^0_n(X)$ if and only if $A \leq_T X^{(n-1)}$.*

Corollary 1.9. *For every $X \subseteq \omega$ and $n \in \omega$, there is $X^{(n)} \in \Sigma^0_n(X) \setminus \Pi^0_n(X)$.*

Definition 1.4. Say that X is **Γ^0_n-complete** for $\Gamma \in \{\Sigma, \Pi\}$ if every $Y \in \Gamma$, $Y \leq_m X$.

By Post's theorem we obtain that $\emptyset^{(n)}$ is Σ^0_n-complete.

1.3 Examples of complete sets

Recall that $(W_e)_{e\in\omega}$ is the canonical enumeration of c.e. sets, i.e., $W_e = \{n : \varphi_e(n)\downarrow\}$.

Proposition 1.10. *The set* $K_1 = \{e : W_e \neq \emptyset\}$ *is* Σ^0_1*-complete.*

Theorem 1.11. *The set* $\mathrm{Fin} = \{e : W_e$ *is finite*$\}$ *is* Σ^0_2*-complete and the set* $\mathrm{Tot} = \{e : W_e = \omega\}$ *is* Π^0_2*-complete.*

Proof. Suppose that $S \in \Sigma^0_2$, then $n \in S$ iff $\exists k \forall m R(n,k,m)$. Define $W_{f(n)}$ via a c.e. approximation as follows. For every s, let k_s be the least k such that $\forall m < s\ R(n,k,m)$ if such k exists and $k_s = s$ otherwise. Suppose we have defined $W_{f(n),s-1}$ and let j_{s-1} be the least element not in $W_{f(n),s-1}$. If $k_s > k_{s-1}$ enumerate j_{s-1} into $W_{f(n),s}$, otherwise $W_{f(n),s} = W_{f(n),s-1}$. We then let $W_{f(n)} = \lim_s W_{f(n),s-1}$. Note that every $W_{f(n),s}$ is computable, uniformly in s and that $W_{f(n),s} \supseteq W_{f(n),s-1}$ is increasing. Thus $m \in W_{f(n)}$ iff $\exists s m \in W_{f(n),s}$ and hence $W_{f(n)}$ is Σ^0_1, so c.e.

If $n \in S$, then $\lim_s k_s \downarrow$ and thus $W_{f(n)}$ is an initial segment of ω. On the other hand if $n \notin S$, then $\lim_s k_s \uparrow$ and furthermore for every m, there is s with $k_s = m$. Thus, $m \in W_{f(n)}$ and $W_{f(n)} = \omega$. □

1.4 Limit lemma

In the proof of Theorem 1.11 we have built a c.e. set as a limit of computable sets. Let us introduce the notion of limit computability more formally. A set A is **limit computable** in X if there is a sequence of finite sets $(F_i)_{i\in\omega}$, computable uniformly in X (i.e., there is an X-computable function f such that for all i,j $f(i,j) = \chi_{F_i}(j)$) such that $\chi_A = \lim_i \chi_{F_i}(j)$.

Theorem 1.12 (Limit lemma)**.** *A set* A *is* $\Delta^0_2(X)$ *(equivalently* $A \leq_T X'$*) if and only if there is a uniformly* X*-computable sequence of finite sets* $(F_i)_{i\in\omega}$ *such that for all* n*,* $A(n) = \lim_i F_i(n)$*.*

Proof. Suppose that $A \leq_T X'$, say $A = \Phi_e^{X'}$. Let X'_s be the stage s approximation of X', i.e., the finite set such that $n \in X'_s$ if and only if $\Phi_n^{X\cap\{0,\dots s\}}(n)\downarrow$. Clearly the sets X'_s are computably uniformly in X. Now let $A_s = \Phi_e^{X'_s}$, note that A_s is finite and computable in X, uniformly in s, as X'_s is a finite oracle (note that for this assumption to hold we not only need that X'_s is finite but also that $X'_s \subseteq \{0,\dots,s\}$). Now, if $\Phi_e^{X'} = i\downarrow$, then there is a shortest $\sigma \prec X'$ such that $\Phi_e^{\sigma}(i)\downarrow$. Furthermore, as X' is c.e., there is a stage s such that for all $t > s$, $X'_t \succeq \sigma$. Thus, for all such t, $A_t(n) = A(n)$.

On the other hand suppose that $A = \lim_s A_s$ with $(A_s)_{s\in\omega}$ uniformly computable in X. Then

$$\begin{aligned} n \in A &\iff \exists m(\forall k > m) n \in A_k \\ n \notin A &\iff \exists m(\forall k > m) n \notin A_k \end{aligned}$$

The first equation shows that A is $\Sigma^0_2(X)$ and the second equation shows that A is $\Pi^0_2(X)$. Thus, A is $\Delta^0_2(X)$. □

2 Cohen 1-generics

We will now start with our discussion of the computability theoretic version of Cohen forcing. The main subject of this section are Cohen 1-generics, to introduce them we do not need a forcing relation yet. The compatible forcing relation will be introduced in the next section.

Cohen invented the forcing method to prove the independence of the continuum hypothesis from ZFC. Since then many variations of forcing exist, and it has established itself as a central tool in mathematical logic. As the name suggests, a *Cohen generic real* $g \in 2^\omega$ is one that is typical and does not have any special properties. The term "special" here can be formalized by

stating that the set of elements having a property P is small in the sense of category, i.e., it is meager. We will discuss this point of view in more detail in a later section.

1-generics were isolated by Jockusch in the 80s even though they appear in many earlier computability theoretic constructions implicitly. We will work with 1-generic elements of 2^ω. However, our definitions and theorems can be easily adapted to ω^ω, and, in fact, any computably presented Polish space.

Definition 2.1. Let $R \subseteq 2^{<\omega}$ be closed upwards, i.e., $\sigma \in R$ implies $\tau \in R$ for all $\tau \succeq \sigma$. We say that $\gamma \in 2^{<\omega}$ **decides** R if

$$\gamma \in R \text{ or } (\forall \sigma \succeq \gamma)\sigma \notin R.$$

An element $g \in 2^\omega$ is **1-generic** if for every upwards-closed c.e. $R \subseteq 2^{<\omega}$, there is $\sigma \prec g$ such that σ decides R.

If R is not upwards-closed then we say that g decides R if it decides its upwards closure.

Definition 2.2. For g 1-generic we say that

1. $\sigma \prec g$ **forces** g **into** $R \subseteq 2^{<\omega}$ if $\sigma \in R$, in which case we also say that g **meets** R.
2. $\sigma \prec g$ **forces** g **out of** $R \subseteq 2^{<\omega}$ if $(\forall \tau \succeq \sigma)\tau \notin R$, in which case we also say that g **avoids** R.

Let us investigate the Turing degrees of 1-generic sets. 1-generics cannot be computable. To see this simply note that for computable $c \in 2^\omega$, the set $R_c = \{\sigma : \sigma \not\prec c\}$ is dense, thus if g is 1-generic, it must be forced into it and cannot be equal to c.

Proposition 2.1. *There is a 1-generic computable from* $\mathbf{0}'$.

Proof. We construct g in stages, i.e., $g \succ \ldots p_s \succ p_{s-1} \succ \ldots p_0$. Fix an enumeration of the c.e. sets $(W_e)_{e\in\omega}$. At stage s, having defined p_{s-1} ask whether there is $q \succeq p_{s-1}$ such that $q \in W_s$. If so, then let $p_s = q$, otherwise $p_s = p_{s-1}$.

Clearly, for every c.e. set W_e, p_e decides W_e. To see that g is indeed an infinite string note that the set $\{\sigma \in 2^{<\omega} : |\sigma| \geq n\}$ is c.e. and dense; thus, at infinitely many stages the length of p_s increases. To see that g is computable from $\mathbf{0}'$, note that at each stage we have to check whether

$$(\exists q \succ p_s)q \in W_e.$$

This is Σ^0_1, so $\mathbf{0}'$ decidable. □

Proposition 2.2. *Suppose that g is 1-generic and W is a c.e. set such that $W \leq_T g$. Then W is computable.*

Proof. Suppose that $W \leq_T g$ and $\Phi^g_e = W$. Consider the set

$$S = \{\sigma : \exists n(\Phi^\sigma_e(n) = 0 \wedge n \in W)\}.$$

This is a c.e. set and thus decided by an initial segment of g. As no initial segment of g can be in S, there is $\sigma \prec g$ such that for every $\tau \succ \sigma$, $\tau \notin S$. We now claim that

$$n \notin W \iff (\exists \tau \succ \sigma)\Phi^\tau_e(n) = 0$$

which would show that W is co-c.e. and thus computable. Certainly, as $g \in [\sigma]$ and $\Phi^g_e = W$, if $n \notin W$, then there is $\tau \succ \sigma$ such that $\Phi^\tau_e(n) = 0$. On the other hand, if $\Phi^\tau_e(n) = 0$ for some $\tau \succ \sigma$, then as $\tau \notin S$ $n \notin W$. Therefore, W is co-c.e., and thus computable. □

Definition 2.3. A degree $\mathbf{d}$ is **generalized low** if $\mathbf{d}' = \mathbf{d} \oplus \mathbf{0}'$ and it is **low** if $\mathbf{d}' = \mathbf{0}'$. Likewise, $x \in 2^\omega$ is (generalized) low if $deg(x)$ is (generalized) low.

Theorem 2.3. *Every 1-generic is generalized low.*

Proof. Consider the sets

$$J_e = \{\sigma : \Phi_e(e)^\sigma \downarrow\}.$$

The sets J_e are c.e. and thus decided by 1-generics, that is if g is 1-generic, then

$$\exists \sigma \prec g (\Phi_e^\sigma(e) \downarrow \vee (\forall \tau \succeq \sigma) \Phi_e(e)^\tau \uparrow) \tag{J_e}$$

Now, for a given string σ, $\emptyset'$ can check whether it satisfies (J_e) and which of the disjuncts it satisfies, i.e., whether it forces into J_e, or out. Thus using $g \oplus \emptyset'$ we can check whether $e \in g'$ by finding the least k such that $g \upharpoonright k$ witnesses (J_e). Thus $g \oplus \emptyset' \geq_T g'$. On the other hand, for any set x, $x' \geq_T \emptyset'$ (see Theorem 1.4), thus the equivalence follows. □

The statements (J_e) in the above proof are often referred to as **forcing the jump** requirements and a set x is said to force the jump if it satisfies all these requirements, i.e., decides all the sets J_e. Unsurprisingly, forcing the jump is a sufficient condition for 1-genericity.

Theorem 2.4. *A real $x \in 2^\omega$ is 1-generic if and only if x forces the jump.*

Proof. The left-to-right direction is trivially true as the sets J_e are c.e.. For the other direction define the computable function $f : \omega \to \omega$ for all n by

$$\Phi_{f(e)}^\sigma(n) = \begin{cases} 1 & \text{if } (\exists \tau \preceq \sigma)\tau \in W_{e,|\sigma|} \\ \uparrow & \text{otherwise} \end{cases}.$$

We then have that $\sigma \in J_{f(e)}$ iff $\sigma \in W_e$ and thus if $\tau \prec x$ decides $J_{f(e)}$ then τ decides W_e, showing that if x forces the jump then it is 1-generic. □

Theorem 2.5 (Friedberg jump inversion)**.** *For every $\mathbf{a} \geq \mathbf{0}'$, there is a 1-generic g with*

$$g' \equiv_T g \oplus \emptyset' \in \mathbf{a}.$$

Proof. We build g in stages interleaving forcing the jump requirements and coding $A \in a$. As usual $p_0 = \emptyset$. Suppose we have define p_{2s}, then let $p_{2s+1} = p_{2s} \frown A(s)$. Then, having defined p_{2s+1} look for the lexicographically least $\sigma \succeq p_{2s+1}$ forcing the jump J_s and let $p_{2s+2} = \sigma$. Let $g = \lim p_s$ and note that $n \in g'$ iff $\Phi_n^{p_{2n+2}}(n) \downarrow$. If $A \geq_T \emptyset'$, then A can check the latter condition and thus can compute g'. On the other hand, we can use $g \oplus \emptyset'$ to compute A inductively. To compute whether $n \in A$, knowing how to compute whether p_{2n-2}, use $\emptyset'$ to determine p_{2n-1} and note that $n \in A$ iff $g(|p_{2n-1}| + 1) = 1$. That $g \oplus \emptyset' \equiv_T g'$ follows from Theorem 2.3 as g forces the jump by construction. □

Corollary 2.6. *The jump $J : 2^\omega \to 2^\omega$ is onto $\{\mathbf{d} \geq \mathbf{0}'\}$.*

2.1 Relative 1-genericity and mutual genericity

Say that $g \in 2^\omega$ is **1-generic relative to** $x \in 2^\omega$, or x-1-generic, if for every upwards-closed x-c.e. $R \subseteq 2^{<\omega}$, g decides R. Furthermore, $x, y \in 2^\omega$ are **mutually 1-generic** if x is y-1-generic, and y is x-1-generic.

Proposition 2.7. *Suppose that g is x-1-generic. Then x is not c.e. in g, unless x is c.e.*

Proof. The proof is quite similar to the proof of Proposition 2.2. Suppose that x is c.e. in g, say $x = W_e^g$ and consider the set

$$S = \{\sigma : \exists n(n \in W_e^\sigma \wedge x(n) = 0\}.$$

S is c.e. in x and thus decided by g. In fact, g avoids S, as otherwise $n \in W_e$ and $x(n) = 0$, a contradiction. Thus, there is $\sigma \prec g$ forcing out of S. We now claim that

$$x(n) = 1 \iff (\exists \tau \succ \sigma) n \in W_e^\tau$$

which shows that x is c.e. If $x(n) = 1$, then as $W_e^g = x$, there is $\tau \succeq \sigma$ such that $n \in W_e^\tau$. On the other hand, i $n \in W_e^\tau$, then as $\tau \notin S$ by the genericity of g, $x(n) = 1$. □